



Hybrid reluctance actuator topology for enhanced linearity and high torque in fast steering mirrors

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ABSTRACT

This paper presents a topology for tilting hybrid reluctance actuators (HRAs) that improves linearity while maintaining high torque output. The primary source of non-linear behavior is identified, and design requirements for the new actuator topology are established using a magnetic equivalent circuit model. Finite element method simulations demonstrate that the proposed design achieves greater linearity and a higher torque to current ratio compared to state of the art actuators. For validation, a fast steering mirror, actuated by the proposed HRA topology, with a $\pm 1^\circ$ range is designed and implemented. Experimental results confirm a threefold reduction in the variation of the torque to current ratio and stiffness, along with a 76% increase in the torque to current ratio compared to the state of the art HRAs.

1. Introduction

In mechatronic systems, the actuation principle is a major contributor to the system performance, particularly in terms of range, bandwidth, and force. High-bandwidth applications typically rely on piezo actuators, which offer high forces but have limited range, while Lorentz actuators offer a larger range but less bandwidth and force density. Hybrid reluctance actuators (HRAs) have demonstrated the highest range-bandwidth product, compared to Lorentz- and piezo-actuated systems, and also obtain forces at equivalent volumes greater than Lorentz actuators [1,2]. Due to high efficiency and compact size, HRAs are widely used across various applications including vibration isolation [3], nanopositioning systems [4,5], deformable mirrors [6], linear scanners [7], and fast tool servos [8,9]. Unlike conventional reluctance actuation, HR actuation is based on a bias flux of a permanent magnet which is superimposed by a coil flux to provide bi-directional forces.

Since fast steering mirrors require high forces, for dynamic scanning and high efficiency to minimize thermal distortions of the mirror surface, HRAs are particularly widespread in fast steering mirrors [10–16]. Fast steering mirrors are optomechatronic systems for scanning or pointing tasks in optical systems. Pointing applications include beam stabilization in free-space optical communication [17], acquisition of optical signals and tracking objects [18], as well as directing laser or light beams [19]. Scanning applications are even broader, ranging from laser scanning [20], material processing [21], scanning optical lithography [22] to confocal microscopy [23]. All these applications

favor a compact fast steering mirror for system integration, which is efficient to minimize the overall system power consumptions and the thermal distortions of the mirror and provides high torque for dynamic scanning and high positioning bandwidth.

However, the state of the art reveals the non-linear system behavior of HRAs, causing a operating point dependent performance and system properties, such as the stiffness, the torque to current ratio and consequently the resonance frequency [13]. Some applications use that non-linear behavior for zero-power gravity compensation [24] or self-holding actuators for mechanical gear shifting [25], but in fast steering mirrors and many other applications a linear system behavior would provide advantages. For vibration isolation and high-precision positioning, zero-stiffness actuation is preferred for its ability to decouple external disturbances. However, the variable negative stiffness of HRAs cannot be compensated independently of the operating point using mechanical flexures, leading to undesirable transmissibility [7, 26]. This variable negative stiffness issue is particularly critical in scanning applications, where the system resonance is tuned to the required scan frequency, and the stiffness variations significantly reduce energy efficiency [27,28]. Furthermore, to compensate that non-linearity by control design, non-linear feedback and/or feedforward compensators [29,30] are required, which requires profound system identification and increases the control complexity. Although advanced air gap shapes can significantly decrease the non-linearity of HRAs, the torque to current ratio is reduced by a factor of 3, compared to state of the art HRAs [31].

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The contribution of this paper is an actuator topology to enhance the linearity of tilting HRAs while maintaining a high torque output. Therefore, the sources of non-linearity are investigated by modeling a one degree of freedom HRA, and a magnetic equivalent circuit to minimize these effects is extracted. Based on this model, a linearized HRA design is developed and verified by FEM simulations. To experimentally evaluate the system behavior, a fast steering mirror prototype, actuated by the linearized HRA, is implemented and tested at various operating points, with detailed analysis of torque-to-current ratio and stiffness variations. Hybrid reluctance actuated fast steering mirrors typically have a range of $\pm 0.2 \dots 3^\circ$, therefore an actuator range of $\pm 1^\circ$ is targeted in this work [1].

2. Torque model of tilting HRAs

A model is developed to represent the fundamental principles of conventional HRAs, incorporating tilt angle α , coil current i_c , and output torque τ . This model is then used to identify the sources of non-linearity and establish the design requirements for the novel actuator topology.

Conventional HRAs consist of a ferromagnetic stator and mover to guide the magnetic fluxes Φ , as illustrated in Fig. 1. Tilting HRAs, such as fast steering mirrors, have a rotational degree of freedom α with the rotation axis nearby the center of the mover. The torque is generated by the reluctance force acting at each side with the lever length l_τ , which is a so called pull-pull configuration since the reluctance force can only apply pulling forces.

The key component of HRAs is a permanent magnet, which introduces the bias fluxes $\Phi_{m1,2}$ on each side. These bias fluxes are superimposed by the coil current i_c dependent flux Φ_c , resulting in an enhanced flux on the left side and a weakened flux on the right. The generated torque can be modeled by [10]

$$\tau(\alpha, i_c) = \frac{(\Phi_{m1}(\alpha) + \Phi_c(i_c))^2 - (\Phi_{m2}(\alpha) - \Phi_c(i_c))^2}{2\mu_0 A_\tau} l_\tau \quad (1)$$

with the cross-section A_τ of each air gap and the permeability of vacuum μ_0 . Since the angular range is limited to $|\alpha| \leq 1^\circ$ in this work, a small angle approximation is used.

For the coil flux, the serially connected air gaps have a length of $l_0 \pm l_\tau \alpha$, resulting in position independent effective air gap length of $2l_0$. By assuming no stray flux and a permeability of $\mu_f \rightarrow \infty$, the coil flux Φ_c is defined by

$$\Phi_c(i_c) = \frac{\mu_0 A_\tau N}{2l_0} i_c \quad (2)$$

with the coil current i_c and the number of turns N . For the center position $\alpha = 0^\circ$, the flux of the permanent magnet Φ_{m0} is divided equally between the left and right air gap, resulting in a bias flux of

$$\Phi_{m1}(\alpha = 0) = \Phi_{m2}(\alpha = 0) = \frac{\Phi_{m0}}{2} \quad (3)$$

With Eqs. (2)–(3), the quadratic flux terms are canceling in Eq. (1), yielding a linear behavior of HRAs in the center position with

$$\tau_0 = \underbrace{\frac{\Phi_{m0} N l_\tau}{2l_0}}_{k_m} i_c \quad (4)$$

This approach shows the linearization of the inherently quadratic reluctance force, which is only valid in the center position. Generally, the bias fluxes $\Phi_{m1,2}$ are depending on the mover position α due to the working air gaps being connected in parallel, resulting in the non-linear behavior of conventional HRAs. Since a reluctance actuator requires a non-zero gradient of the working reluctances to generate a force, the bias fluxes of the permanent magnet cannot be constant for variable mover positions [31].

The initial step to achieve a linear HRA topology across the full angular range is to reapply the previous method used to linearize the

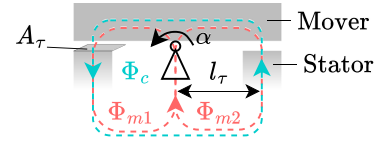


Fig. 1. Principle of a HRA with the parallel bias flux paths of the permanent magnet denoted as $\Phi_{m1,2}$ and the coil flux as Φ_c .

HRA at its center position. By assuming bias fluxes that in-/decrease linearly depending on the mover position

$$\Phi_{m1,2} = \frac{\Phi_{m0}}{2} \pm k_\phi \alpha, \quad (5)$$

the quadratic terms α^2 and i_c^2 , as well as the mixed-terms αi_c , are canceling in Eq. (1) with Eq. (2). This results in an exact linearization of the torque

$$\tau_{lin} = \frac{\Phi_{m0} l_\tau (\Phi_c + k_\phi \alpha)}{\mu_0 A_\tau} = \underbrace{\frac{\Phi_{m0} k_\phi l_\tau}{\mu_0 A_\tau}}_{\kappa} \alpha + k_m i_c \quad (6)$$

with the rotational stiffness κ and the torque to current ratio, or so called motor-constant, k_m defined in Eq. (4). Hence, a new actuator topology is needed to establish a linearly related bias flux and achieve the desired linear behavior of the HRA.

3. Magnetic equivalent circuit model

In order to identify a linear HRA topology that ensures a linearly dependent bias flux, a magnetic equivalent circuit is used as it provides a simple and accurate modeling approach [32]. The extracted magnetic equivalent circuit is subsequently verified by finite element method simulations and evaluated by experiments.

The magnetic equivalent circuit comprises the magneto motive forces Ψ , which models the sources of the bias and coil flux Φ , and the reluctances R of the HRA, as shown in Fig. 2. The working reluctances $R_{1,2}(\alpha)$ demonstrate the left and right air gap and carry the fluxes $\Phi_{1,2}$. These fluxes are the main contributors of torque generation, and consists of the superimposition of the coil and bias flux. By modifying the air gap shape, these working reluctance can be adjusted, so that the non-linearity of the HRA is decreased [31]. This promising approach showed a significantly reduced torque to current ratio compared to the conventional HRA structure at equivalent size. As the conventional HRA design guides the magnetic flux in the working air gaps to be mainly tangential to the rotation axis, as depicted in Fig. 1, yielding the higher torque to current ratio. Since the torque to current ratio is desired to be as high as possible, the working reluctances are adopted from the conventional design. By approximating $\sin \alpha \approx \alpha$ for the targeted range of $|\alpha| \leq 1^\circ$, the working reluctances are defined by

$$R_{1,2}(\alpha) = \frac{l_{eff1,2}}{\mu_0 A_\tau} \quad (7)$$

$$l_{eff1,2} = l_0 \mp l_\tau \alpha,$$

with the air gap length in the center position l_0 , the cross-section A_τ , the distance to the rotation axis l_τ and the permeability of vacuum μ_0 . The magneto motive forces of the symmetric actuation coils are defined by

$$\Psi_{c1,2}(i_c) = \frac{N i_c}{2} \quad (8)$$

with the total number of turns N and the coil current i_c .

To obtain the desired linearly dependent bias flux in each air gap, two bias flux sources are introduced. The general idea is to adapt the magneto motive forces to avoid the non-linear behavior of the bias fluxes in conventional HRA designs. Therefore, the magneto motive

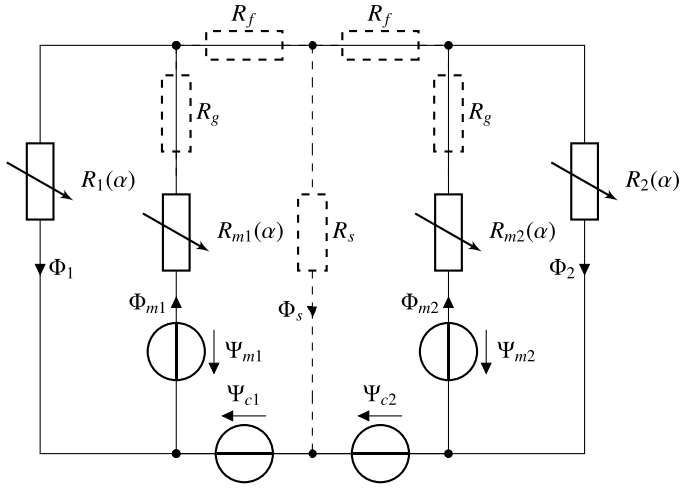


Fig. 2. Magnetic equivalent circuit of the linearized HRA with the reluctances R , the magneto motive forces Ψ and the fluxes Φ . The extended magnetic equivalent circuit is depicted by the dashed reluctances.

forces and their internal resistances are adjusted according to the air gap length of the working reluctances, resulting to

$$\begin{aligned} \Psi_{m1,2}(\alpha) &= H_c l_{\text{eff}1,2} \\ R_{m1,2}(\alpha) &= \frac{l_{\text{eff}1,2}}{\mu_0 A_m} \end{aligned} \quad (9)$$

with the coercivity H_c , the cross-section of the permanent magnets A_m and the effective permanent magnet length l_{eff} . Due to the parallel connection of both working reluctances and the bias flux sources, the coupling of the magnetic equivalent circuit needs to be considered to verify the linearization effect of this approach. By applying Gauss's and Ampere's law on the magnetic equivalent circuit in Fig. 2, without the dashed reluctances R_s , R_f and R_g , the following circuit equations are derived

$$\begin{aligned} \Psi_{m1,2} &= \Phi_{m1,2} R_{m1,2} + \Phi_{1,2} R_{1,2} \\ \Psi_{c1} + \Psi_{c2} &= \Phi_1 R_1 - \Phi_2 R_2 \\ \Phi_{m1} + \Phi_{m2} &= \Phi_1 + \Phi_2. \end{aligned} \quad (10)$$

Based on the magnetic equivalent circuit and the extracted Eqs. (7) - (10), the bias fluxes $\Phi_{m1,2}$ show the required linear behavior and the unknown parameters from Eq. (5) are identified to be

$$\begin{aligned} \Phi_{m0} &= \frac{2H_c A_m \mu_0 A_r}{A_r + A_m} \\ k_\Phi &= \frac{l_r}{l_0} \Phi_{m0} \end{aligned} \quad (11)$$

with static bias flux in the center position Φ_{m0} and gradient of the bias flux k_Φ . The torque of the linearized HRA can be rearranged to

$$\tau_{lin}(\alpha, i_c) = \underbrace{\frac{\Phi_{m0}^2 l_r^2}{2l_0 \mu_0 A_r}}_{\kappa} \alpha + \underbrace{\Phi_{m0} \frac{N l_r}{2l_0}}_{k_m} i_c \quad (12)$$

showing that the rotational stiffness κ of the linearized HRA has a quadratic relation to the static bias flux Φ_{m0} and that the torque to current ratio k_m is linearly related to the static bias flux Φ_{m0} .

The proposed magnetic equivalent circuit enables a linear HRA, however, unintended reluctance must also be taken into account to obtain an accurate model. Therefore, the magnetic equivalent circuit is extended with the stray reluctance R_s to model the stray flux of the actuator, the non-working air gap reluctance R_g and the reluctance of the ferromagnetic core R_f . With the extended magnetic equivalent

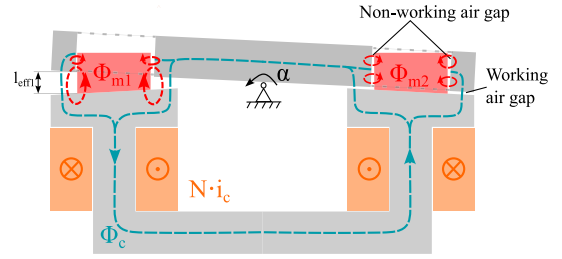


Fig. 3. Principle of the linearized HRA with the variable effective length permanent magnets for bias flux adaption. The permanent magnets are plunged into the ferromagnetic mover, which partially short-circuits the upper part via the non-working air gap.

circuit, the exact linear property of the circuit is not maintained. So the reluctances R_f and R_g needs to be kept as low as possible and the stray reluctance R_s need to be as high as possible, for high linearity. These design considerations and an actuator topology, which linearly adapts the magneto motive forces $\Psi_{m1,2}$, as well as, the respective internal resistances $R_{m1,2}$ to the mover position is required to obtain the linearized HRA.

4. Linearized actuator topology

To achieve an actuator topology that closely resembles the magnetic equivalent circuit, a permanent magnet with a effectively changing length is introduced on each side of the actuator, as shown in Fig. 3. The structure that enables the effective length change is a ferromagnetic mover with holes for permanent magnets. The ferromagnetic mover partially short-circuits the length of the immersed part of the permanent magnets, thus changing its effective length. This mechanism provides the required variable magneto motive force and internal resistance, defined in Eq. (9). For example for a negative mover deflection α , the effective permanent magnet length $l_{\text{eff}1}$ is increased and $l_{\text{eff}2}$ is reduced by $l_r \alpha$. To reduce torques introduced by the non-working air gaps between the mover and the permanent magnets, the air gap is chosen to be orthogonal to the working air gaps. Additionally, to achieve a low reluctance of R_g , the air gap length needs to be as low as possible. By the placement of the permanent magnets directly at the working air gaps, it is also expected to reduce the stray flux, since the permanent magnet is fully embraced by the ferromagnetic mover. Furthermore, the state of the art HRA design with one permanent magnet conventionally requires a ferromagnetic structures in the center to guide the bias flux, as shown in Fig. 4(c). The mover surfaces at the working air gaps are oriented radially to the rotation axis, ensuring that the magnetic fluxes primarily follow a tangential path relative to the axis. This design is expected to achieve a high torque to current ratio, comparable to that of conventional HRA designs. Furthermore, the stator surfaces in the working air gaps are tilted by the maximum deflection of $\pm 1^\circ$, to compensate for the reduced air gap on the inner side of the actuator and to ensure a uniformly distributed flux density at maximum deflection.

5. Simulative linearity and performance verification

To verify the linearized HRA topology, finite element method simulations (Maxwell 3D, Ansys Inc., USA) are conducted, incorporating non-linear BH curves for the ferromagnetic components (layered 0.3 mm, carbon steel sheets, Grade 1015) and the permanent magnets (NdFeB, Grade N45). The mover torque is evaluated in relation to the angular deflection α and the coil current i_c in a range of ± 1 A.

The cross-section of the CAD-model, used for the linearized HRA simulations, is depicted in Fig. 4(a). The stator consists of two series-connected coils with a total number of turns $N = 300$ and two

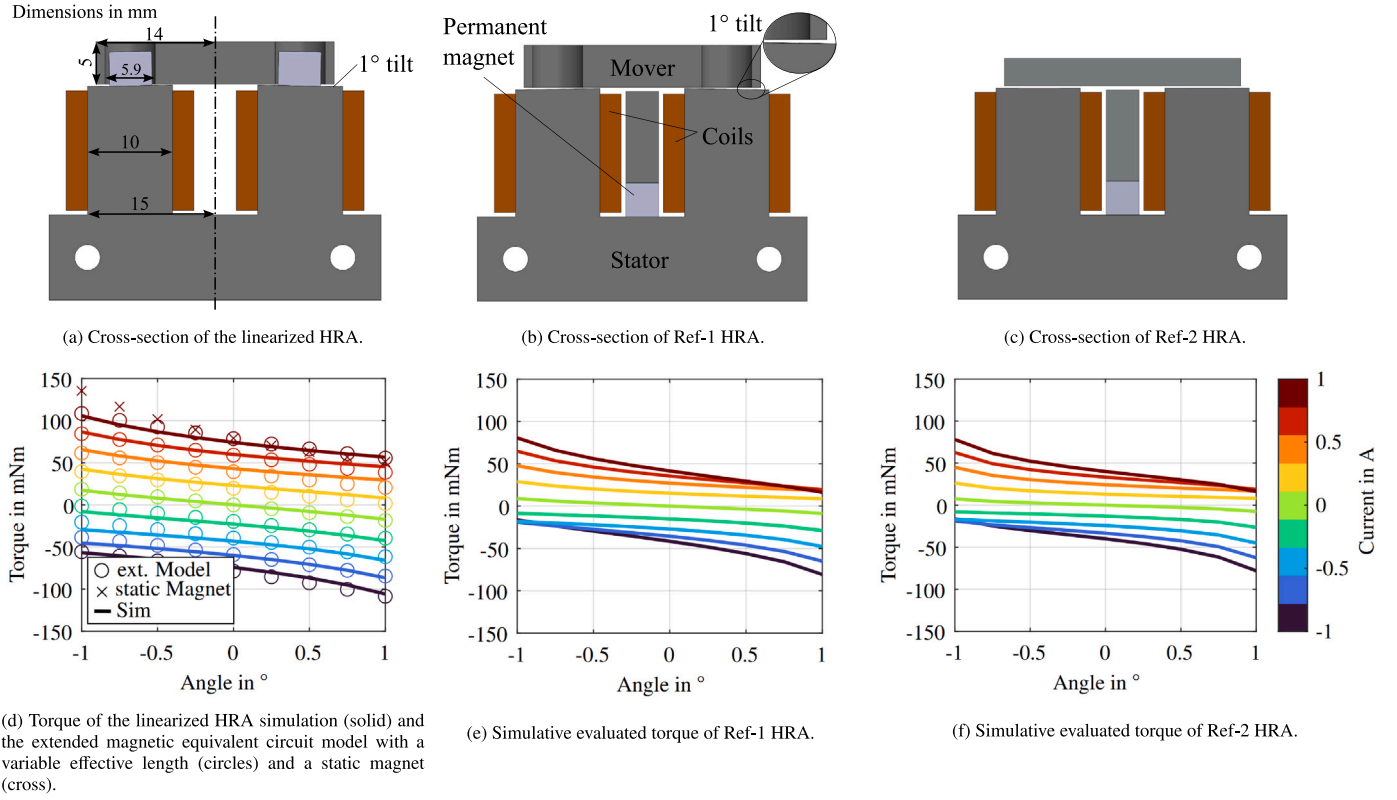


Fig. 4. The linearized HRA and the reference HRAs Ref-1,2 used for the simulative linearity and performance verification. Finite element method simulations are performed for the represented designs and the resulting torques for various coil currents with respect to the mover deflection are shown.

cylindrical permanent magnets with a height of 4 mm and a diameter of 5 mm, resulting in cross-section of $A_m = 19.6 \text{ mm}^2$, respectively. In order to verify the performance with respect to linearity and torque output of the linearized HRA, two reference HRAs are additionally simulated with equivalent coils and an equivalent ferromagnetic stator structure. In contrast to the linearized HRA, a permanent magnet with an identical height of 4 mm and a cross-section of $2A_m$ ($3.92 \times 10 \text{ mm}^2$) is used as a reference design, since it shows the best performance in terms of range-bandwidth product in the literature [1,13]. The first reference HRA Ref-1 has an equivalent mover with holes, to enable a comparison of the differing permanent magnet locations. The reluctances of the working air gaps are kept unchanged, as shown in Fig. 4(b). Since the mover shape with holes increases the working reluctances for the conventional design, a second reference HRA Ref-2 with a solid mover is investigated, as shown in Fig. 4(c). For representing the state of the art HRA, the stator surface at the working air gaps are not tilted. To achieve equivalent angular accelerations at the same torque of each design, the mover height of the Ref-2 is matched to the inertia of the linearized HRA design. The last parameter to ensure a fair comparison is the length of the non-working air gap, which is defined to $450 \mu\text{m}$ for all designs.

Fig. 4(d) shows the simulation results for the linearized HRA design with solid lines and the modeled torque of the extended magnetic equivalent circuit with the parasitic reluctances R_f and R_s identified by simulations and R_g is defined by the dimensions of the non-working air gap. The modeled torque matches the simulation result with a root mean square error of 0.4 mNm , verifying the introduced extended magnetic equivalent circuit. To demonstrate the influence of permanent magnets with variable length and to distinguish the improvement resulting from reduced stray flux, the extended circuit is evaluated using a fixed magnet length at a coil current of 1 A , as shown in Fig. 4(d). The results confirm the expected improvement in linearity purely by the variable magnet length. For high coil currents, partial saturation

Table 1

Stiffness and the torque to current ratio of the fitted linear model and the corresponding maximum variations, identified by simulations.

Parameters	lin. HRA	Ref-1	Ref-2
k_{m0} in mNm A^{-1}	81.1	48.7	46.1
κ_0 in mNm^{-1}	-19.1	-17.5	-15.8
$\max(k_m/k_{m0})$ in %	10.2	27.6	42.8
$\max(\kappa/\kappa_0)$ in %	175	729	1133

of the ferromagnetic material is observable by the reduced distance of the torque curves at the center position. This material induced non-linearity is not considered in the model, therefore the ferromagnetic parts need to be operated in their linear region. Furthermore, the non-zero reluctances R_f , R_g and the stray reluctance R_s introduce a minor non-linear behavior, observable towards larger deflections α .

Both reference designs obtain higher non-linearity of the torque output, observable at the change of the slopes for variable coil currents, as well as the highly varying distances of the torque curves in Figs. 4(e) and 4(f). To enable a quantitative comparison of the HRA designs, a linear model with the stiffness κ_0 and the torque to current ratio k_{m0} is fitted with a least-squares approach to the simulated torque. The resulting parameters of each linear model and the maximum variation of the stiffness and the torque to current ratio are shown in Table 1. First of all the linearized HRA shows the highest torque to current ratio of the investigated designs. This may be caused by the reduced stray reluctance R_s , since the linearized HRA introduces the bias flux as close as possible to the working air gaps and no flux guiding structure in the center is required. Due to the lower stray flux, the bias flux is increased at equivalent volumes of the permanent magnets, resulting in an enhanced torque to current ratio. Additionally, the negative stiffness of the linearized HRA is higher, which is both caused by the higher static bias flux Φ_{m0} , according to Eq. (12). The maximum relative variation of

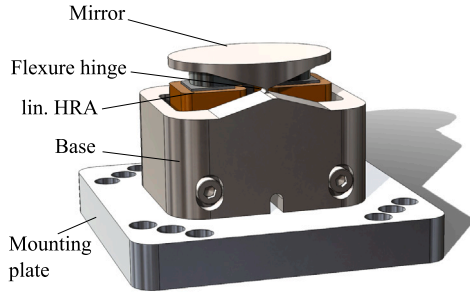


Fig. 5. FSM design comprising the linearized HRA, a monolithic flexure design with a elliptical 1×1.5 inch mirror and a mounting plate.

the torque to current ratio is for the linearized HRA design with about 10% the lowest, verifying that the linearity is significantly improved compared to the reference HRAs. The maximum stiffness change of the linearized HRA with 175% represents a significant improvement in linearity compared to the reference designs, especially to the state of the art HRA (Ref-2). Lastly, the positive impact of the tilted stator surfaces on linearity is evident in the reduced variation of stiffness and the current-to-torque ratio, as shown in Table 1, confirming this design considerations for the linearized HRA.

6. Fast steering mirror design

A fast steering mirror actuated by the linearized HRA is designed to verify the enhanced system behavior in an experimental setup, as shown in Fig. 5. In order to limit the mover motion to one degree of freedom up to high frequencies and to compensate the negative stiffness of the linearized HRA to stabilize the mover in its center position, a flexure hinge is required.

The flexure hinge is implemented by a notch design for having comparably low stiffness in the rotational degree of freedom and a highest possible stiffness in the remaining degrees of freedom, since it is commonly used in high performance fast steering mirrors [11,13]. The rotation axis is defined by the thinnest part of the hinge, which is in this design chosen to $400 \mu\text{m}$. Ti-6Al-4V is used for the flexure hinge due to relatively high yield strength to avoid damage of the flexure hinge at high deflection angles and a relatively low density for low inertia of the moving part. The remaining dimensions of the hinge are tuned to shift the undesired mode shapes to frequencies of $>4 \text{ kHz}$. Finite element method simulation revealed a rotational stiffness of $\kappa_f = 115 \text{ mNm}^\circ^{-1}$, compensating the negative stiffness of the linearized HRA in full range according to the simulation results in Table 1. In contrast, the stiffness variations of the reference HRA designs are too high to be compensated by the flexure in full range, since a stiffer flexure would exceed the torque and current limits of actuator designs. This leads to open-loop unstable operating regions towards higher deflection angles α and coil currents i_c .

To minimize the number of mechanical parts of the fast steering mirror, a monolithic design, that combines the housing, the flexure hinge and the mirror is used. The elliptical mirror has a clear aperture of 1×1.5 inch for 45° mounting proposes and the reflectivity is achieved by polishing the material, which is sufficient to identify the system behavior. Inside, the mover of the linearized HRA is mounted to the upper part of the fast steering mirror. The stator is mounted with two screws to the base and the elongated holes enable the compensation of manufacturing tolerances for a precise air gap length of $l_0 = 175 \mu\text{m}$ in the center. An aluminum plate is connected to the base to provide the several mounting options on a 40 and 50 mm grid.

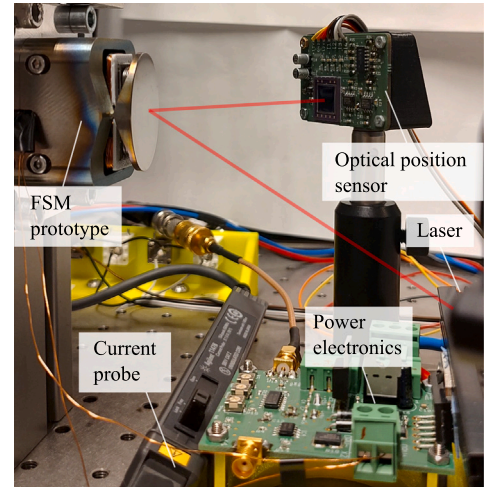


Fig. 6. Experimental setup with the laser beam deflected by the fast steering mirror (FSM) prototype and a optical position sensor to measure the deflection angle α .

7. Experimental setup

For experiments a fast steering mirror prototype is built and a setup with a laser, an optical position sensor and power electronics to drive the coil current is implemented, as shown in Fig. 6. The power electronics is supplied with $\pm 30 \text{ V}$ to drive current through the series-connected coils and to compensate the dynamics of the electric subsystem, an analog PI control with a -3 dB -bandwidth of 20 kHz is integrated.

A 0.9 mW laser with a wavelength of 635 nm is pointed towards the center of the fast steering mirror and the reflected laser beam is deflected on the sensor-plane. As an optical position sensor system, a two dimensional position-sensitive detector (Hamamatsu, S5991) with transimpedance amplifiers is used. The resulting angular deflection is evaluated by

$$\alpha = \frac{1}{2} \arctan \frac{y_{\text{PSD}}}{l_s} \quad (13)$$

with the vertical position on the position-sensitive detector y_{PSD} and the sensing distance $l_s = 70 \text{ mm}$. Additionally, the division by 2 transforms the optical deflection into the mechanical deflection angle α of the fast steering mirror.

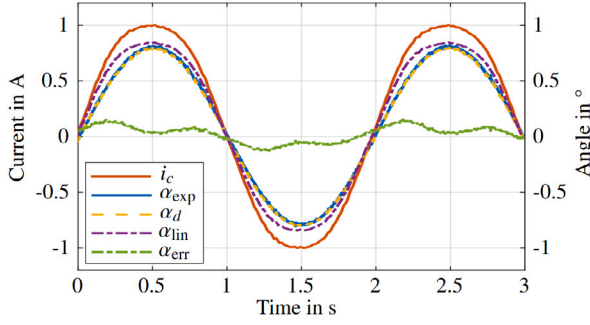
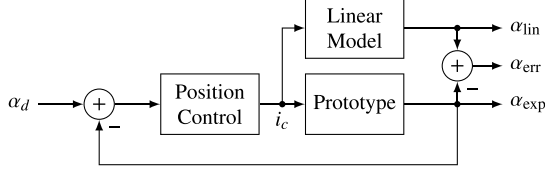
As rapid prototyping system, a dSPACE MicroLab Box with a sampling frequency of 100 kHz is used to perform the A/D conversion and process the position sensing data. In addition, to enable the system identification in various offset deflections α with coincident amplitudes, a cascaded PID-control is tuned in a loop-shaping approach to achieve a -3 dB -position-bandwidth of 2 kHz [33].

Initial tests of the fast steering mirror obtained a range of $\pm 0.95^\circ$, being equivalent to an air gap length tolerance of only $8.7 \mu\text{m}$. Additionally, the fast steering mirror can be open-loop operated in full range, therefore the flexure stiffness must be higher than the negative stiffness of the linearized HRA $\kappa_f > -\kappa$.

8. Performance and linearity evaluation

In order to evaluate the stiffness κ of the linearized HRA and the torque to current ratio k_m at various operating points of α and i_c , the system dynamics are modeled in the Laplace domain by a mass-spring-damper system with the time delay T_d of the digital prototyping system

$$P(s) = \frac{\alpha(s)}{i_c(s)} = \frac{k_m(\alpha, i_c)}{Js^2 + Ds + \kappa(\alpha, i_c) + \kappa_f} e^{-sT_d} \quad (14)$$

(a) System response for the desired trajectory α_d 

(b) Block diagram of the system with the position control, the linear model and the prototype of the fast steering mirror.

Fig. 7. Experimental results of a sinusoidal scan with an amplitude of 0.8° , showing the measured coil current i_c , the measured mover deflection α_{exp} , the simulated deflection α_{lin} and model deflection error α_{err} .

with the moment of inertia from the CAD-Model $J = 1.2 \text{ kg m}^2$, the flexure stiffness k_f defined in Section 6, and the damping coefficient D , which is experimentally identified with the subsequent measurements.

8.1. Large scan amplitude

The first experiment is performed at low frequencies to identify the system behavior for large amplitudes of the coil current and the deflection angle. Therefore, a sinusoidal desired trajectory α_d of 0.8° and a frequency of 0.5 Hz is applied to the position control input. The corresponding coil current is measured with a current probe (Agilent, 1147 A) and the system response is measured by the optical position sensor, shown in Fig. 7(a). Since, the position control adapts the coil current to follow desired scan trajectories, the non-linearity of the HRA is observable by the distortions of the coil current. To demonstrate the non-linearity in comparison to an exact linear system and to verify the finite element method simulations, the dynamic model from Eq. (14) is evaluated with the parameters (k_0 , k_{m0}) of the fitted linear model from the simulation results Table 1. The corresponding system response α_{lin} is simulated with the measured coil current as input, as shown in the block diagram in Fig. 7(b). The error between linear model and the experimental results is denoted as α_{err} . The linear model response is steeper around the center and towards the maximum deflection it is flatter than the experimental system response, resulting in a model error of $\alpha_{err} = 0.03^\circ$ at $t = 0.5 \text{ s}$. This non-linearity corresponds to the simulation results, that show an increasing negative stiffness and torque to current ratio towards maximum deflection in the second and fourth quadrant of Fig. 4(d). The low-frequent deflection and the flexure stiffness limit the operating range to the second and fourth quadrants of the torque output. This experiment demonstrates that the linearized HRA largely exhibits the expected behavior in terms of stiffness and torque to current ratio for large scan amplitudes, confirming the simulation results.

8.2. System dynamics at offset positions

To thoroughly evaluate changes in stiffness and the torque to current ratio, the fast steering mirror is examined at various offset positions α , in a closed-loop manner with a small signal amplitude of 0.01° .

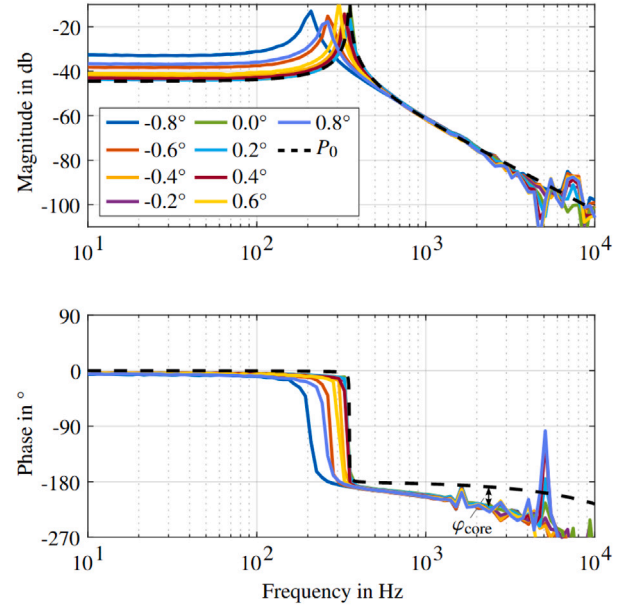


Fig. 8. Experimental results of the system dynamics in various offset positions.

Fig. 8 shows the dynamic system response of the fast steering mirror prototype. The experiment verifies the expected behavior with a positive stiffness for low frequencies, observable by the spring line with the mover deflection in phase to the coil current, followed by a resonance and a -40 dB/dec mass-line for higher frequencies with the mover deflection out of phase to the coil current. At $f > 4 \text{ kHz}$, structural modes of the flexure hinge are observable. The stiffness is reduced towards larger offset deflections, resulting in the shift of the resonance frequency towards lower frequencies. By fitting the dynamic model from Eq. (14) to the measurement results, the stiffness of the linearized HRA κ and the torque to current ratio k_m are evaluated for each operating point with P_0 demonstrating the dynamic model for the center position. The comparison of the modeled and measured system response shows a higher phase lag due to the core losses φ_{core} , which are not considered in the dynamic model [7,34]. Asymmetries in stiffness variation are observable at positive and negative offset positions. These deviations from the expected symmetrical behavior may originate from tolerances in the left and right working air gaps or variations in the magnetization of the permanent magnets used in the prototype.

To evaluate the maximum variation in stiffness and torque to current ratio, the dynamic models for the minimum values at the center position and the maximum values at the 0.8° offset position are related. The experimental results compared to the simulated maximum variations of the linearized HRA, as well as, the maximum variations of the Ref-1 and Ref-2 designs are shown in Fig. 9. The experimental stiffness variation obtains a maximum of 380% , which is about twice the simulated variation in Table 1. A possible effect, that could be the origin of these variations could be the hysteresis of the mover core, being partially magnetized by the offset current, causing a magneto motive force that increases the pulling force towards the stator yoke. To prove this hypothesis, a prototype based on a core material with lower coercivity is required, which is the goal of future work. Although the stiffness variation is experimentally higher than expected, the variation is reduced by a factor of 3 compared to the simulated state of the art design (Ref-2). The torque to current ratio variation stays below the simulated variation of 10% from Table 1, confirming the reduction by a factor of 4, compared to the state of the art (Ref-2). With the cross-over frequency chosen to be in the frequency range of the mass-line, these low variations in the system response result in low variation of the

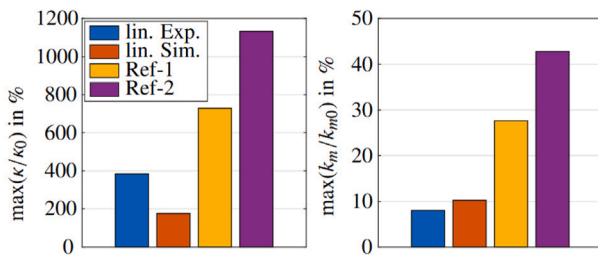


Fig. 9. Maximum variation of the actuator stiffness κ/κ_0 and the torque to current ratio k_m/k_{m0} . A comparison of the simulated/experimental linearized HRA (lin. Exp./Sim.) and the reference HRAs (Ref-1/2).

cross-over frequency, which is desirable regarding control robustness and position-independent system performance.

In summary, the proposed topology of the linearized HRA design enhances the linearity of hybrid reluctance actuated fast steering mirrors, while providing higher torque output than the state of the art.

9. Conclusion

This work describes the fundamental principle of HRAs based on a bias flux and a coil flux. By assuming a linear relation between the bias flux and the mover deflection for each working air gap, an exact linear torque model, featuring an operating point independent torque to current ratio k_m and stiffness κ . Based a magnetic equivalent circuit with two magneto motive forces for the bias flux that adapt their forces corresponding to the air gap length, the desired linear bias flux relation is achieved. A permanent magnet that changes its effective length by immersing into the ferromagnetic mover enables the adaptation of the magneto motive force. This linearized HRA actuator topology approximates the required magnetic behavior. Finite element method simulations verify the extended magnetic equivalent circuit and reveal an enhanced linearity, as well as, a higher torque of the linearized HRA in comparison to state of the art. Experimental results verify the maximum variation of 10 % for the torque to current ratio k_m/k_{m0} and for the stiffness variation κ/κ_0 a value of 380 %, which is a reduction by a factor of 3, compared to the state of the art, while improving the maximum torque by 76 %. However, the simulation underestimated the stiffness variation by approximately 50 %, potentially due to core effects.

Future research will focus on advanced magnetic core materials to reduce core effects and decrease reluctance R_f . Alternatively, the focus could be on advanced permanent magnet shapes to minimize R_g , further improving linearity and extending the approach to two degrees of freedom for broader applications.

CRediT authorship contribution statement

Alexander Pechhacker: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christoph Palka:** Investigation, Data curation. **Ernst Csencsics:** Writing – review & editing, Supervision. **Georg Schitter:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alexander Pechhacker reports financial support was provided by Micro-Epsilon Messtechnik GmbH & Co KG. Alexander Pechhacker has patent #Reluktanz-Aktor, Fast Steering Mirror und Verfahren, 4401/P/1085-DE pending to MICRO-EPSILON-MESSTECHNIK GmbH & Co. K.G. If

there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] Csencsics E, Schitter G. Exploring the Pareto fronts of actuation technologies for high performance mechatronic systems. *IEEE/ASME Trans Mechatronics* 2021-04;26(2):1053–63. <http://dx.doi.org/10.1109/tmech.2020.3016087>.
- [2] Cigarini F, Ito S, Troppmair S, Schitter G. Comparative finite element analysis of a voice coil actuator and a hybrid reluctance actuator. *IEEE J Ind Appl* 2019-03;8(2):192–9. <http://dx.doi.org/10.1541/ieejia.8.192>.
- [3] Takahashi K, Makino R, Ito S. Vibration isolator using hybrid reluctance actuator toward quasi-zero stiffness. *IFAC-PapersOnLine* 2023;56(2):3392–7. <http://dx.doi.org/10.1016/j.ifacol.2023.10.1487>.
- [4] Chen L, Niu Y, Yang X, Zhu W-L, Zhu L-M, Zhu Z. A novel compliant nanopositioning stage driven by a normal-stressed electromagnetic actuator. *IEEE Trans Autom Sci Eng* 2022-10;19(4):3039–48. <http://dx.doi.org/10.1109/tase.2021.3105683>.
- [5] Du X, Shen D, Huang P, Zhu L, Zhu Z. Dual-stage fast tool servo cascading a primary normal-stressed electromagnetic stage with a secondary piezo-actuated stage. *Precis Eng* 2023-03;80:171–9. <http://dx.doi.org/10.1016/j.precisioneng.2022.12.005>.
- [6] Bowers-Rubin R, Hinz P, Kuiper S, Dillon D. Performance of large-format deformable mirrors constructed with hybrid variable reluctance actuators II: initial lab results from FLASH. In: Shaskan SB, Ruane GJ, editors. *Techniques and instrumentation for detection of exoplanets x*. SPIE; 2021. <http://dx.doi.org/10.1117/12.2593535>.
- [7] Ito S, Cigarini F, Schitter G. Flux-controlled hybrid reluctance actuator for high-precision scanning motion. *IEEE Trans Ind Electron* 2020-11;67(11):9593–600. <http://dx.doi.org/10.1109/tie.2019.2952829>.
- [8] Wu D, Xie X, Zhou S. Design of a normal stress electromagnetic fast linear actuator. *IEEE Trans Magn* 2010;46(4):1007–14. <http://dx.doi.org/10.1109/tmag.2009.2036606>.
- [9] Huang W-W, Zhu Z, Zhang X, Zhu L-M. A hybrid electromagnetic-piezoelectric actuated tri-axial fast tool servo integrated with a three-dimensional elliptical vibration generator. *Precis Eng* 2024-03;86:213–24. <http://dx.doi.org/10.1016/j.precisioneng.2023.12.006>.
- [10] Kluk DJ, Boulet MT, Trumper DL. A high-bandwidth, high-precision, two-axis steering mirror with moving iron actuator. *Mechatronics* 2012-04;22(3):257–70. <http://dx.doi.org/10.1016/j.mechatronics.2012.01.008>.
- [11] Kuiper S, Crowcombe W, Human J, Dekker B, Nieuwkoop E, Meskers A, Gert. High-bandwidth and compact fine steering mirror development for laser communications. *Tech. Rep.*, 2017, the.
- [12] Rüddenklau R, Sinn A, Schitter G. Comparison of high performance hybrid variable reluctance fast steering mirrors. In: 2024 12th international conference on control, mechatronics and automation. IEEE; 2024, p. 367–73. <http://dx.doi.org/10.1109/iccma63715.2024.10843930>.
- [13] Csencsics E, Schlarp J, Schopf T, Schitter G. Compact high performance hybrid reluctance actuated fast steering mirror system. *Mechatronics* 2019-10;62:102251. <http://dx.doi.org/10.1016/j.mechatronics.2019.102251>.
- [14] Csencsics E, Schlarp J, Schitter G. High-performance hybrid-reluctance-force-based tip/tilt system: Design, control, and evaluation. *IEEE/ASME Trans Mechatronics* 2018-10;23(5):2494–502. <http://dx.doi.org/10.1109/tmech.2018.2866272>.
- [15] Long Y, Mo J, Chen X, Liang Q, Shang Y, Wang S. Design of a rotary electromagnetic actuator with linear torque output for fast steering mirror. *J Magn* 2015;20:69–78. <http://dx.doi.org/10.4283/JMAG.2015.20.1.069>.
- [16] Zhang W, Li W, Zheng X, Yu W, Huang H. Modeling and designing a hybrid reluctance actuator for fast-steering mirror. *Opt Eng*, Bellingham 2022;61(05). <http://dx.doi.org/10.1117/1.OE.61.5.054106>.
- [17] Mokbel HF, Yuan W, Ying LQ, Hua CG, Roshdy AA. Research on the mechanical design of two-axis fast steering mirror for optical beam guidance. In: *Proceedings of the 1st international conference on mechanical engineering and material science*. Atlantis Press; 2012. <http://dx.doi.org/10.2991/mems.2012.137>.
- [18] Guelman M, Kogan A, Kazarian A, Livne A, Orenstein M, Michalik H, et al. Acquisition and pointing control for inter-satellite laser communications. *IEEE Trans Aerosp Electron Syst* 2004-10;40(4):1239–48. <http://dx.doi.org/10.1109/taes.2004.1386877>.
- [19] Xiang S, Wang P, Chen S, Wu X, Xiao D, Zheng X. The research of a novel single mirror 2D laser scanner. In: *SPIE proceedings*, vol. 7382, 2009-07, p. 73821A. <http://dx.doi.org/10.1117/12.832651>.

- [20] Hafez M, Sidler T, Salathé R, Jansen G, Compter J. Design, simulations and experimental investigations of a compact single mirror tip/tilt laser scanner. *Mechatronics* 2000;10(7):741–60. [http://dx.doi.org/10.1016/s0957-4158\(99\)00093-8](http://dx.doi.org/10.1016/s0957-4158(99)00093-8).
- [21] Hedding LR, Lewis RA. Fast steering mirror design and performance for stabilization and single-axis scanning. In: *SPIE proceedings*. 1990-09, p. 14–24. <http://dx.doi.org/10.1117/12.21547>.
- [22] Zhou Q, Ben-Tzvi P, Fan D, Goldenberg A. Design of fast steering mirror systems for precision laser beams steering. In: *2008 international workshop on robotic and sensors environments*. IEEE; 2008-10, <http://dx.doi.org/10.1109/rose.2008.4669196>.
- [23] Yoo HW, van Royen ME, van Cappellen WA, Houtsmuller AB, Verhaegen M, Schitter G. Automated spherical aberration correction in scanning confocal microscopy. *Rev Sci Instrum* 2014;12;85(12). <http://dx.doi.org/10.1063/1.4904370>.
- [24] Stadler G, Csencsics E, Ito S, Schitter G. High precision hybrid reluctance actuator with integrated orientation independent zero power gravity compensation. *IEEE Trans Ind Electron* 2022;12;69(12):13296–304. <http://dx.doi.org/10.1109/tie.2021.3137444>.
- [25] Immonen P, Ruuskanen V, Pyrhönen J. Moving magnet linear actuator with self-holding functionality. *IET Electr Syst Transp* 2018-09;8(3):182–7. <http://dx.doi.org/10.1049/iet-est.2017.0079>.
- [26] Zhu T, Cazzolato B, Robertson WS, Zander A. Vibration isolation using six degree-of-freedom quasi-zero stiffness magnetic levitation. *J Sound Vib* 2015;358:48–73. <http://dx.doi.org/10.1016/j.jsv.2015.07.013>.
- [27] Schlarp J, Csencsics E, Dobliger G, Schitter G. Design of a mechanical tunable resonant fast steering mirror. In: *2020 IEEE/ASME international conference on advanced intelligent mechatronics (AIM)*, Boston, USA (virtual conference), July 6–9, 2020. IEEE; 2020, <http://dx.doi.org/10.1109/AIM43001.2020.9159032>.
- [28] Ito S, Troppmair S, Lindner B, Cigarini F, Schitter G. Long-range fast nanopositioner using nonlinearities of hybrid reluctance actuator for energy efficiency. *IEEE Trans Ind Electron* 2019;66(4):3051–9. <http://dx.doi.org/10.1109/tie.2018.2842735>.
- [29] Chen L, Liu Q, Zhu L, Zhu Z. On the actuation nonlinearity of normal-stressed electromagnetic nanopositioning stages. *Precis Eng* 2024-01. <http://dx.doi.org/10.1016/j.precisioneng.2024.01.009>.
- [30] Katalenic A, de Boeij J, Butler H, van den Bosch P. Linearization of a current-driven reluctance actuator with hysteresis compensation. *Mechatronics* 2013-03;23(2):163–71. <http://dx.doi.org/10.1016/j.mechatronics.2013.01.004>.
- [31] Pechhacker A, Csencsics E, Schitter G. Decreasing the non-linearity of hybrid reluctance actuators by air gap design. *Precis Eng* 2024;91:255–62. <http://dx.doi.org/10.1016/j.precisioneng.2024.09.018>.
- [32] Forstner G, Kugi A, Kemmetmüller W. A magnetic equivalent circuit based modeling framework for electric motors applied to a PMSM with winding short circuit. *IEEE Trans Power Electron* 2020;35(11):12285–95. <http://dx.doi.org/10.1109/TPEL.2020.2986042>.
- [33] Schmidt RM, Schitter G, Rankers A, van Eijk J. *The design of high performance mechatronics : high-tech functionality by multidisciplinary system integration*. Amsterdam: Delft University Press; 2020.
- [34] Pechhacker A, Csencsics E, Schitter G. Enhanced energy efficiency of hybrid reluctance actuators by bias flux adaptation. In: *Proceedings of international conference on mechatronics*. 2025, <http://dx.doi.org/10.1109/ICM62621.2025.10934938>.



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